

MethylCentroid Distributions and Sufficient Statistics

MethylPipeline

1 Overview

MethylCentroid aggregates per-position methylation across a cohort of samples and stores *sufficient statistics* that allow parameter estimation for multiple distributions without retaining the full sample matrix. This enables:

- memory-efficient centroid construction and storage,
- incremental updates (add/remove samples),
- GPU-friendly batch computation.

The centroid is implemented as a `MethylExtendedCentroid` and can be compared with `MethylCentroidPair` under multiple distributional models.

2 Notation

For a genomic position, let the sample index be $i = 1, \dots, N$. Let mC_i and uC_i be methylated and unmethylated counts, respectively. Define:

$$\begin{aligned} n_i &= mC_i + uC_i \quad (\text{coverage}) \\ x_i &= \frac{mC_i}{n_i} \quad (\text{methylation fraction}) \end{aligned}$$

The centroid stores aggregated statistics across the contributing samples.

3 Stored Sufficient Statistics

For each position, the centroid stores:

$$\begin{aligned} N &= \sum_i 1 \\ S_x &= \sum_i x_i \\ S_{x^2} &= \sum_i x_i^2 \\ \log X &= \sum_i \log(x_i) \\ \log(1 - X) &= \sum_i \log(1 - x_i) \\ \Sigma n &= \sum_i n_i \\ \Sigma n^2 &= \sum_i n_i^2 \\ \Sigma mC &= \sum_i mC_i \\ \Sigma uC &= \sum_i uC_i \\ \Sigma mC^2 &= \sum_i mC_i^2 \\ \Sigma uC^2 &= \sum_i uC_i^2 \\ S_{x^3} &= \sum_i x_i^3 \\ S_{x^4} &= \sum_i x_i^4 \\ c_0 &= \sum_i \mathbb{I}[x_i = 0] \\ c_1 &= \sum_i \mathbb{I}[x_i = 1] \end{aligned}$$

The first set (through $\log(1 - X)$) is always present in the extended centroid. The remaining statistics are optional and are enabled by the extended builder for downstream distribution selection and modeling.

4 Supported Distributions

4.1 Normal Distribution (Approximation)

For large-sample regimes, methylation fractions can be approximated as:

$$x_i \sim \mathcal{N}(\mu, \sigma^2).$$

The centroid provides:

$$\hat{\mu} = \frac{S_x}{N}$$

$$\hat{\sigma}^2 = \frac{S_{x^2} - \frac{S_x^2}{N}}{\max(N-1, 1)}.$$

These estimates are stable and require only (N, S_x, S_{x^2}) .

4.2 Beta Distribution

Methylation fractions are naturally bounded in $[0, 1]$, making the Beta distribution a strong default model:

$$x_i \sim \text{Beta}(\alpha, \beta).$$

The log-likelihood depends on:

$$\sum_i \log(x_i) \quad \text{and} \quad \sum_i \log(1 - x_i).$$

Thus, $(N, \log X, \log(1 - X))$ are sufficient for maximum-likelihood estimation. The centroid computes $\hat{\alpha}, \hat{\beta}$ via a bounded MLE routine and provides derived moments:

$$\mathbb{E}[x] = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(x) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

4.3 Beta-Binomial Distribution

Counts are modeled as:

$$mC_i \mid p_i \sim \text{Binomial}(n_i, p_i), \quad p_i \sim \text{Beta}(\alpha, \beta).$$

This captures over-dispersion in coverage. The centroid stores aggregated count statistics $(\Sigma mC, \Sigma uC, \Sigma n, \Sigma n^2)$ to approximate coverage mean and variance and to inform model selection. In practice, the comparison engine uses aggregated counts for LRT-style tests and pooled (α, β) from the log-sum statistics.

4.4 Beta Mixture Model (BMM)

A flexible multi-modal model:

$$x_i \sim \sum_{k=1}^K w_k \text{Beta}(\alpha_k, \beta_k).$$

Fitting mixtures requires sample-level values or a histogram. To avoid massive centroid sizes, binned statistics are *optional*:

- **Binned stats (optional):** per-position histograms (bin edges + counts)
- **Masking:** the BMM is typically fitted only on a subset of significant positions

The resulting mixture parameters are stored in a separate `MethylBetaMixtureCentroid`, with an optional mask to limit positions.

5 Why Sufficient Statistics

Storing sufficient statistics rather than full sample matrices yields:

- **Memory savings:** per-position aggregates scale linearly with positions, not with positions $\times$ samples.
- **Streaming updates:** new samples update sums without reloading old samples.
- **GPU efficiency:** vectorized aggregates map well to GPU kernels.
- **Model flexibility:** different distributions can be fit from the same centroid.

6 Practical Notes

- **Base centroid:** $(N, S_x, S_{x^2}, \log X, \log(1 - X))$ are always present.
- **Extended stats:** $\Sigma n, \Sigma n^2, \Sigma mC, \Sigma uC$, etc. support Beta-Binomial selection and higher-moment diagnostics.
- **BMM histograms:** only stored when explicitly enabled; otherwise the refinement stage can fall back to sample-based values or masked subsets.

7 Distribution Choice in Comparisons

MethylCentroidPair supports per-position distribution selection (`auto`, `beta`, `normal`, `beta_binomial`, `beta_mixture`). The chosen distribution determines:

- the p-value test,
- how *mean* and *overlap* are computed (configurable),
- which positions are eligible for mixture refinement.

8 Summary Table

Distribution	Parameters	Required Stats
Normal	μ, σ^2	N, S_x, S_{x^2}
Beta	α, β	$N, \log X, \log(1 - X)$
Beta-Binomial	α, β	$N, \log X, \log(1 - X), \Sigma n, \Sigma mC$
Beta Mixture	$\{w_k, \alpha_k, \beta_k\}$	Histogram or samples (optional bins)